GRAPHYTI
Start

OUR TEAM:

Howell Chan

Cofounder | AI/Unity Developer

- REP (Spec. in CS) @ NTU
- Work experience in AI & full-stack engineering
- Former NTU researcher in Computer Graphics

Eunice Lee

Cofounder | Graphics/Unity Developer

- REP (Spec. in CS) @ NTU
- Work experience in Computer Graphics
- Former NTU researcher in Computer Graphics

WHAT IS GRAPHYTI:

A platform that generates and redesigns infinite 3D worlds in seconds to train AI systems of the future

WHAT ARE INTERACTIVE WORLD MODELS...

Meet Google's newest interactive world model, Genie 3...

BESIDES BIG BRAND NAMES, STARTUPS ARE IN THE RACE TO CREATE THE WORLD'S MOST POWERFUL SPATIAL INTELLIGENCE...

Google: Genie 3

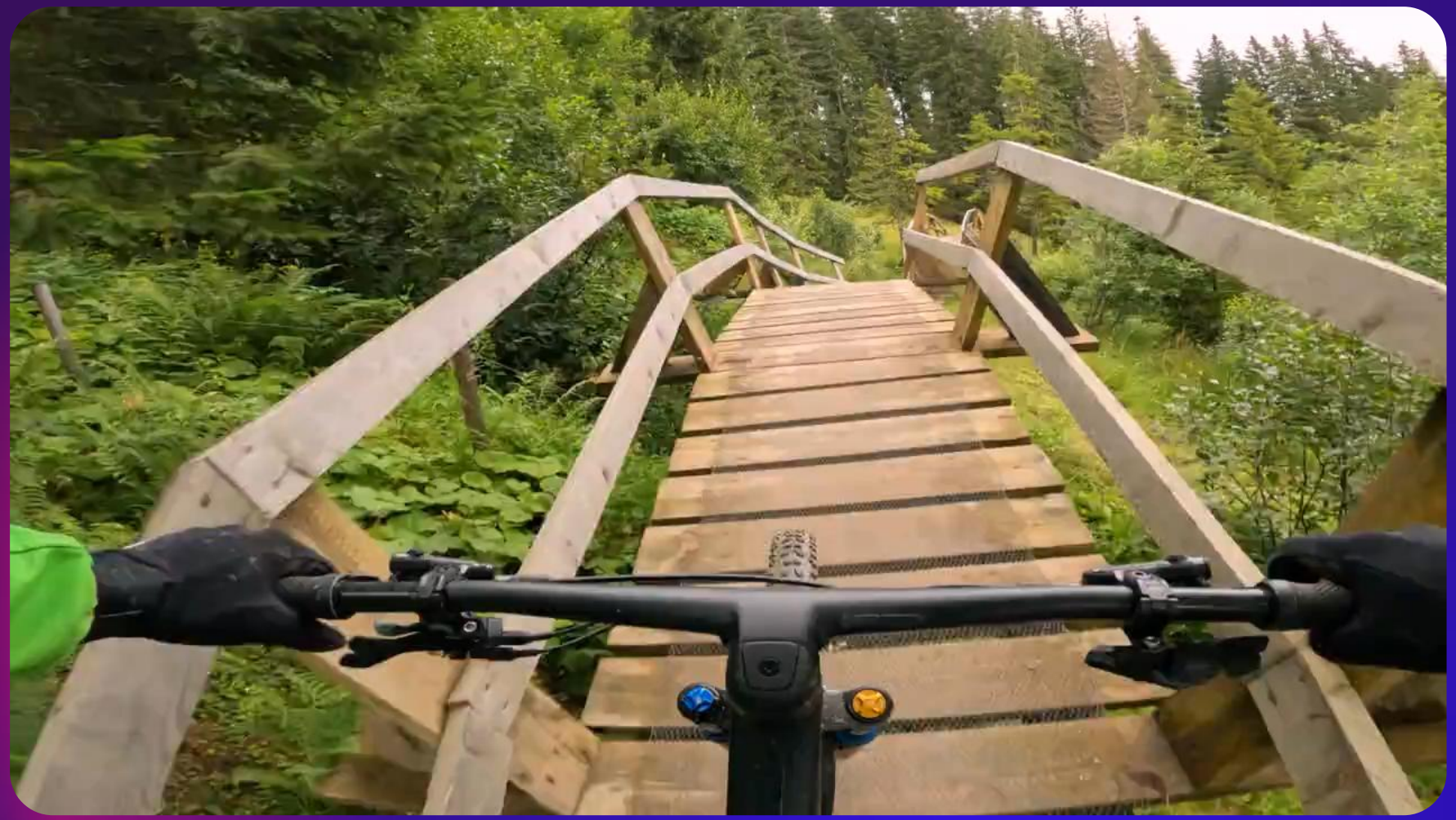
Tencent: HunyuanWorld

Microsoft : WHAMM

Odyssey AI

World Labs

WHAT DO WE NEED TO TRAIN SUCH SPATIAL INTELLIGENCE...



Currently, training data for world models and video models comes from footage captured in difficult or hazardous real-world environments

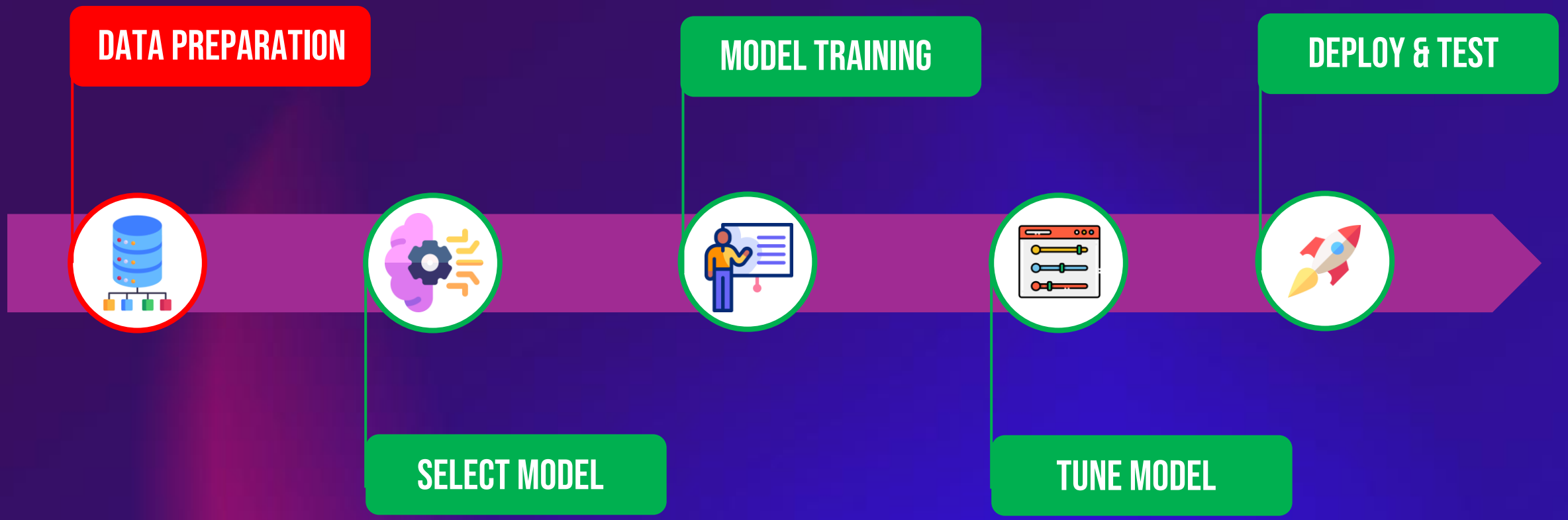
WHAT DO WE NEED TO TRAIN SUCH SPATIAL INTELLIGENCE...

Snowboarding

Skydiving

Training world models require unprecedented costly manual labour and highly calibrated cameras, along with extensive footage

THE CURRENT WORKFLOW FOR TRAINING AI MODELS...



SUPPORTED BY INTERVIEWS FROM...

World Labs

HTX

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

ZEISS

THE 1ST STEP THAT MANY AI COMPANIES STRUGGLE WITH

CURRENTLY, THERE ARE LIMITED SOLUTIONS TO EASILY GENERATE DATASETS

CURRENTLY, THERE ARE COUNTLESS OPEN-SOURCE , NO-CODE SOLUTIONS & EVEN APIS FOR AI MODEL INFERENCE

81% OF COMPANIES STILL STRUGGLE WITH AI DATA QUALITY, PUTTING ROI OF AI INVESTMENTS AT RISK

THE GLOBAL SYNTHETIC DATA GENERATION MARKET IS US$218M IN 2023 & PROJECTED TO REACH US$1.8B BY 2030

HENCE, WE'RE FOCUSING ON **DATA PREPARATION**

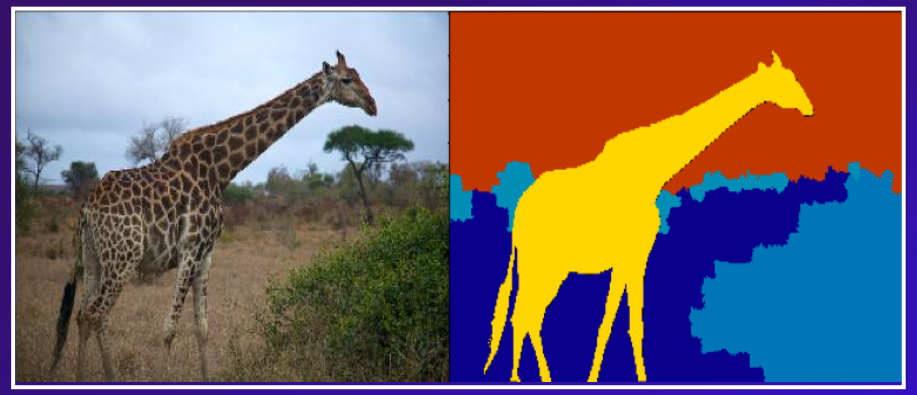
SOURCE: QLIK, GVR

WHY THE CURRENT WORKFLOW FOR DATASET CREATION IS PROBLEMATIC...

1. MANUAL COLLECTION

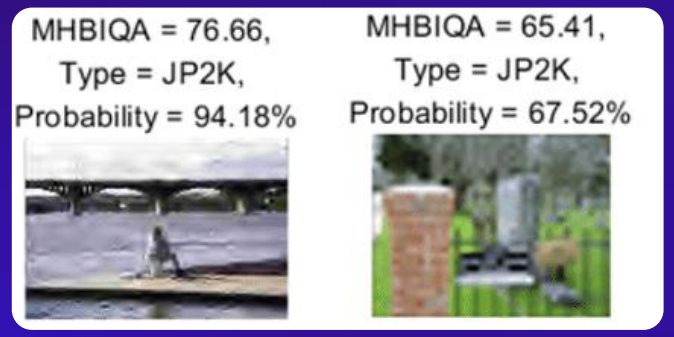
- **Advanced Camera Systems** for high-quality environmental capture
- **Online Images/Videos** (Films, Art, Drone Footage, YouTube Videos, Stock Images/Videos)

2. CURATE, PREPROCESS & ANNOTATE



- **Programming Required for More Sophisticated Preprocessing**
- **Manual Annotation**

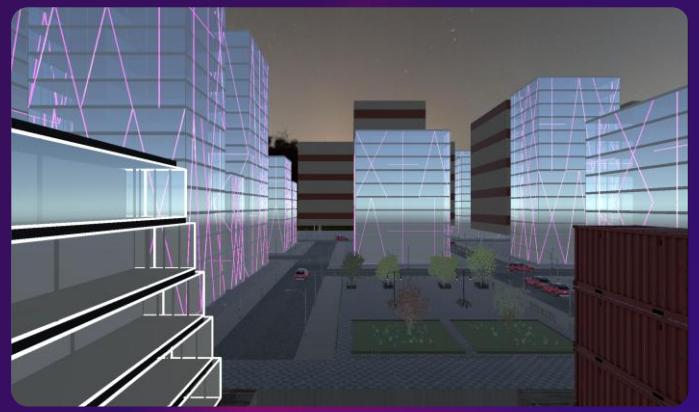
3. DATA QA



- **Benchmark models on real validation sets**
- **Human-in-the-loop** assessments

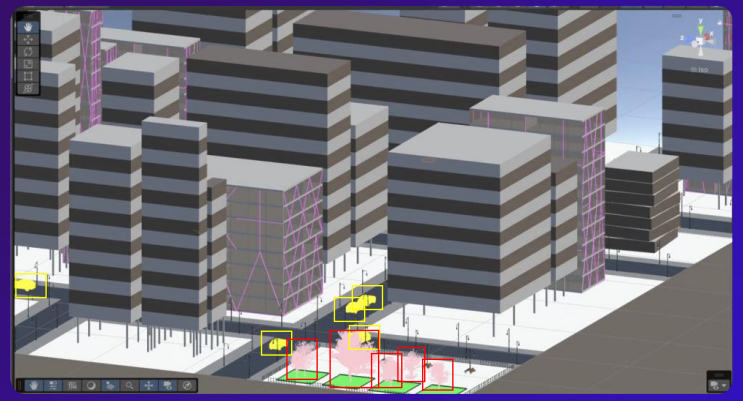
INTRODUCING OUR 3D SYNTHETIC DATA GENERATION PLATFORM, GRAPHYTI

1. 3D SCENE GENERATION



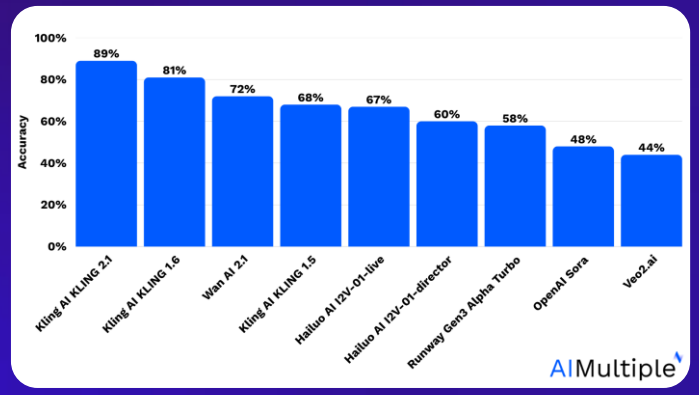
- **Large-Scale Custom Photorealistic 3D Scene Generation via Text Prompts**
- **Easy Capture of Image/Video Datasets with camera flythrough functions**

2. AUTOMATED ANNOTATIONS



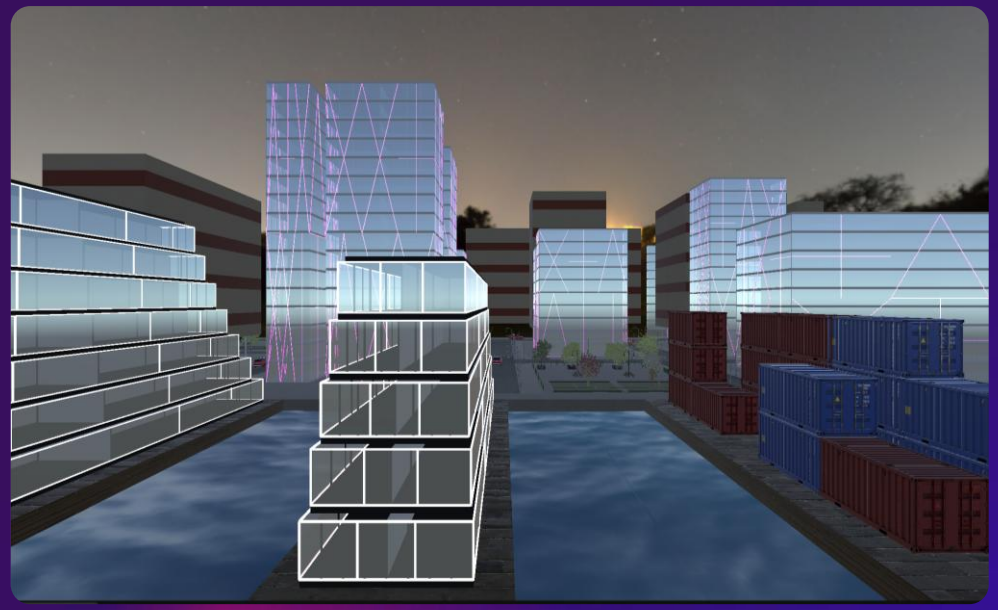
- **Auto Data cleaning & Restructuring** into train/val/test splits
- **Autogenerated Annotations:** Bounding boxes/polygons, segmentation masks, pose landmarks, scene captions

3. AUTOMATED TESTING



- **Auto Benchmark models on real validation sets**
- **Human-in-the-loop** assessments

FEATURES: ALGORITHM-BASED 3D SCENE GENERATION



Minimal AI Approach

- Not using AI images to train AI
- True 3D Geometry

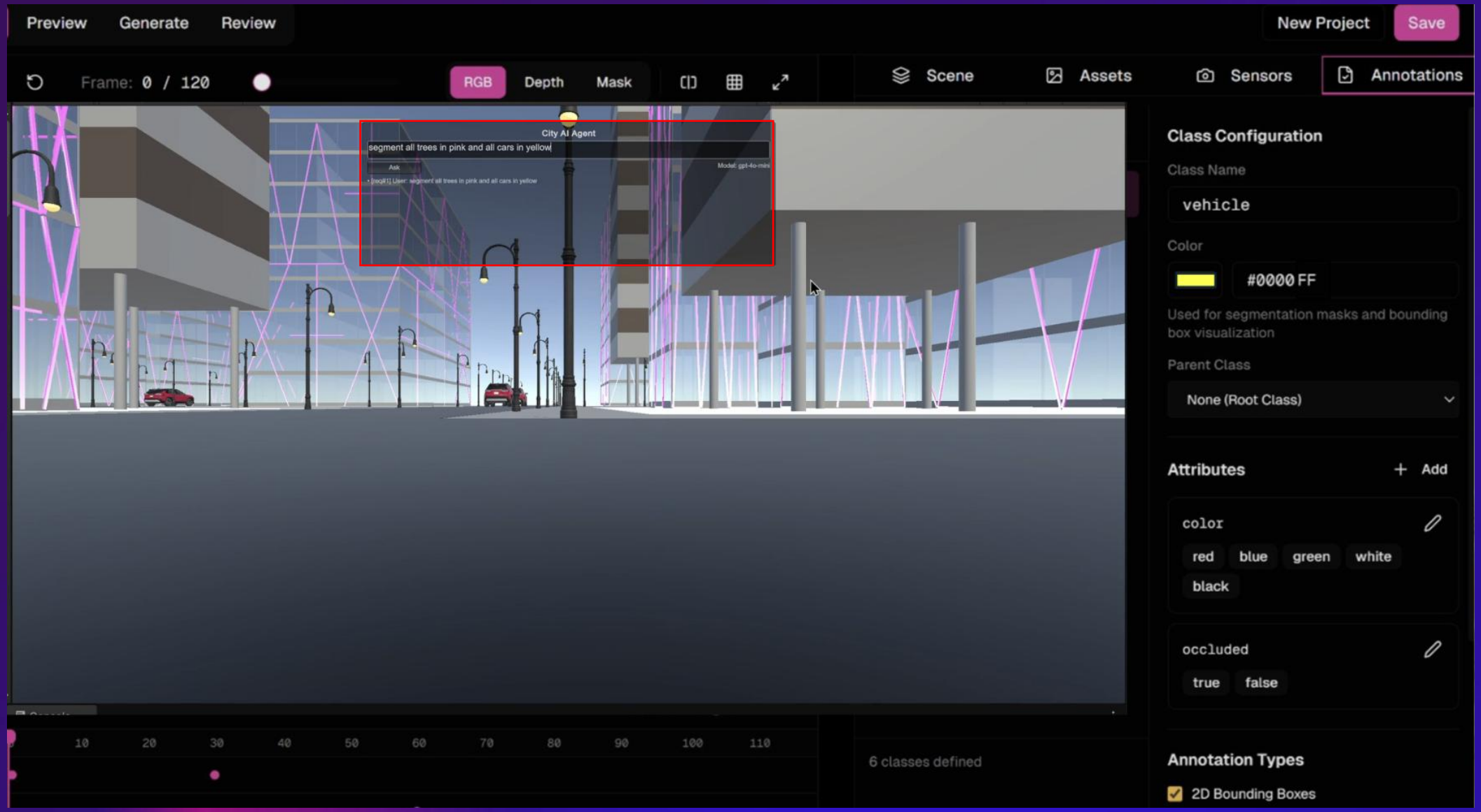
Procedural/Algorithmic Approach

- Layouts, Assets controlled by Algorithms
- Reproducible & Controllable → No Black Box

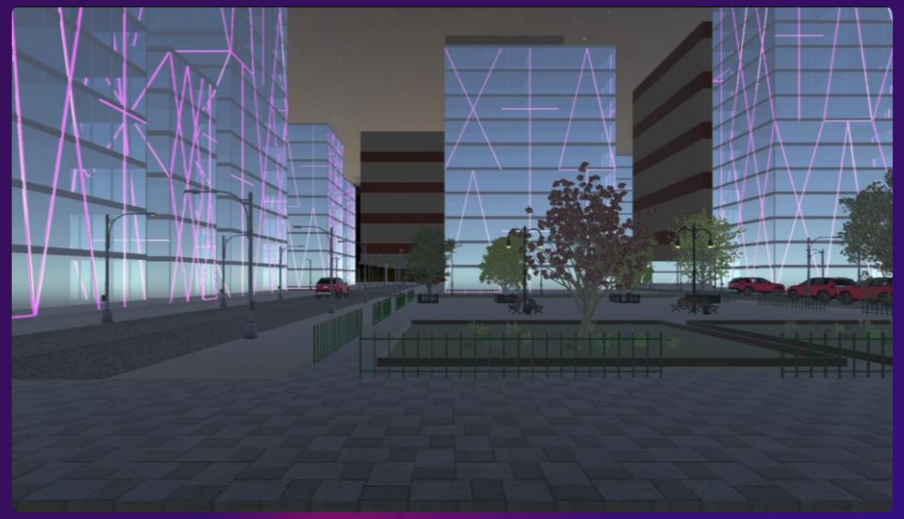
AI as a Design Co-Pilot, Not a Creator

- LLM to interpret natural language prompts
- Translate into algorithmic parameters for 3D scene generation

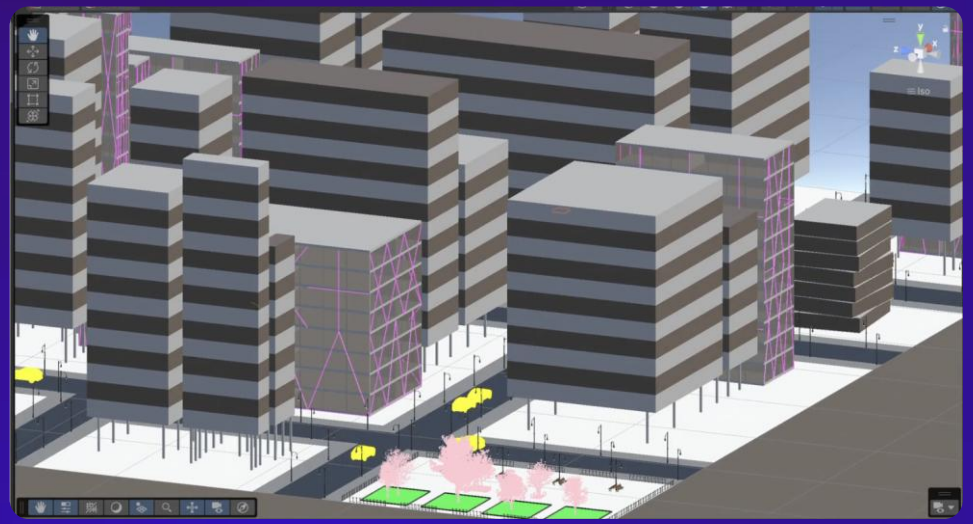
FEATURES: AI DESIGN CO-PILOT



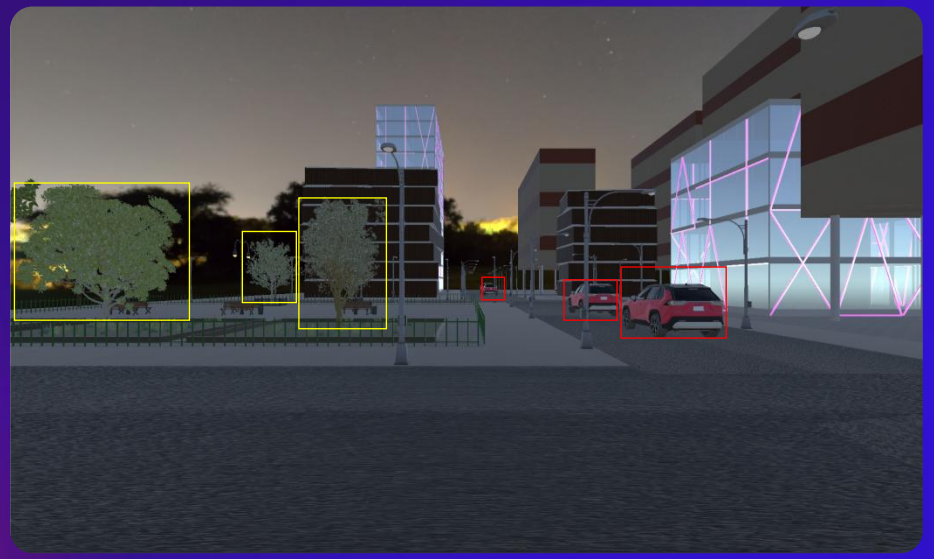
FEATURES: MULTIMODAL DATA OUTPUTS



Raw RGB Image/Video



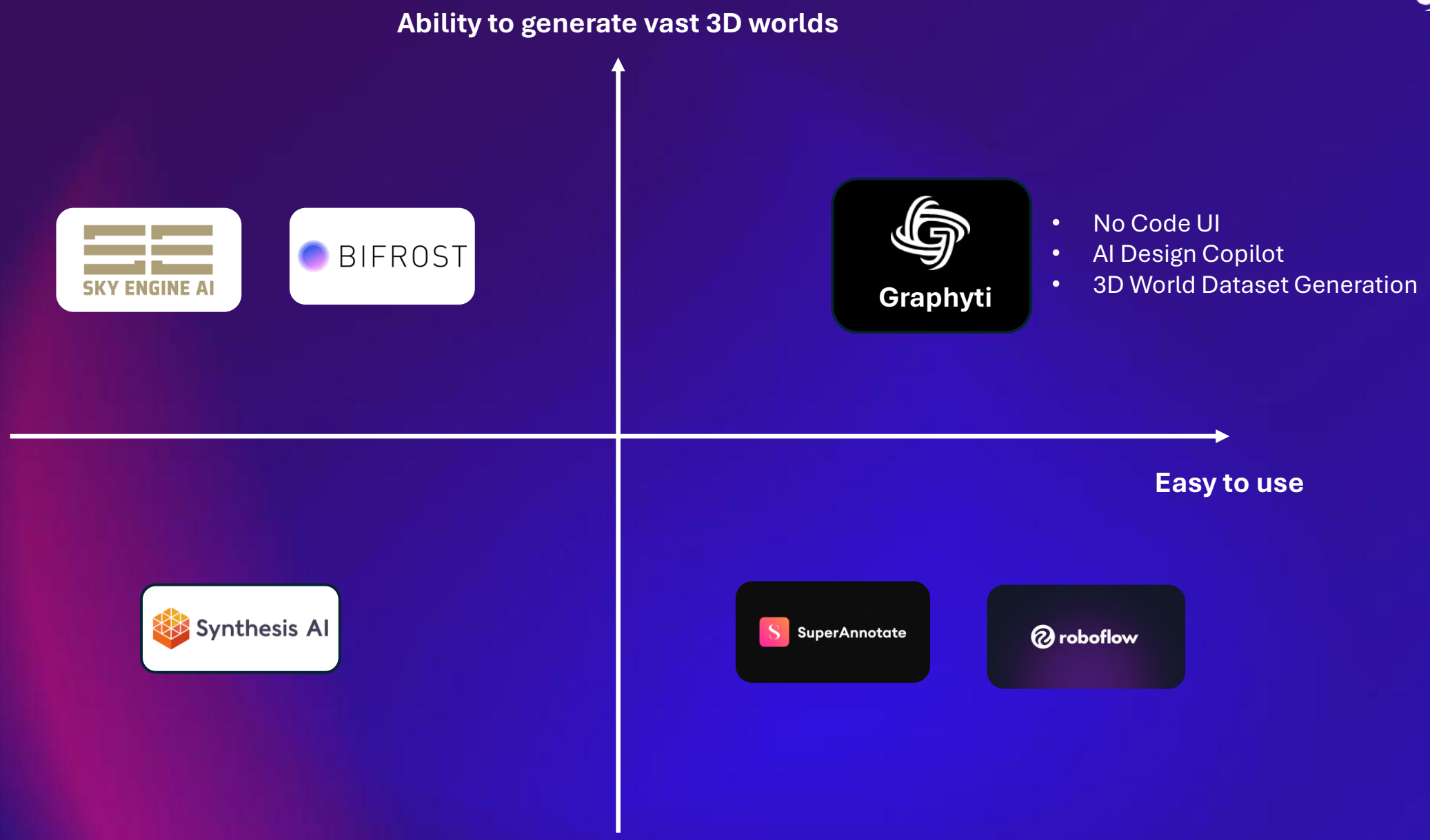
Segmentation Masks



Bounding Boxes

etc...

COMPETITOR ANALYSIS



OUR BUSINESS MODEL

TARGET CUSTOMERS

AI/ML teams building computer vision & world models

Research Labs needing reproducible datasets for experimentation

Enterprises in autonomous vehicles, robotics, VR/AR, gaming & simulation

REVENUE STREAMS

Subscription Tiers

- **Free/Basic:** Limited scenes, small dataset export
- **Pro:** Higher fidelity, large-scale dataset generation, advanced annotations
- **Enterprise:** Unlimited usage, custom integrations, private deployments

Custom Services

- Tailored dataset generation for enterprises & research labs
- API integrations for industry-specific pipelines (e.g. robotics, AV)

MILESTONES

Port63 Challenge – Web3.0 Champion

Algorand Global Hackathon: 2nd Runners Up

Entered Algorand Incubator

Prior to Aug 2024

Aug 2024

Sep 2024

Nov 2024

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Completed Proof-of-Concept for NTU Research Project

Microsoft

Proud to partner with Microsoft for Startups

Entered Microsoft Startups Founders Hub programme

DEEP TECH PIONEERS hello tomorrow

Emerged as top 850/4500+ teams worldwide in Hello Tomorrow